

Physics A (H156, H556)

Exploring - Nuclear fission and fusion ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

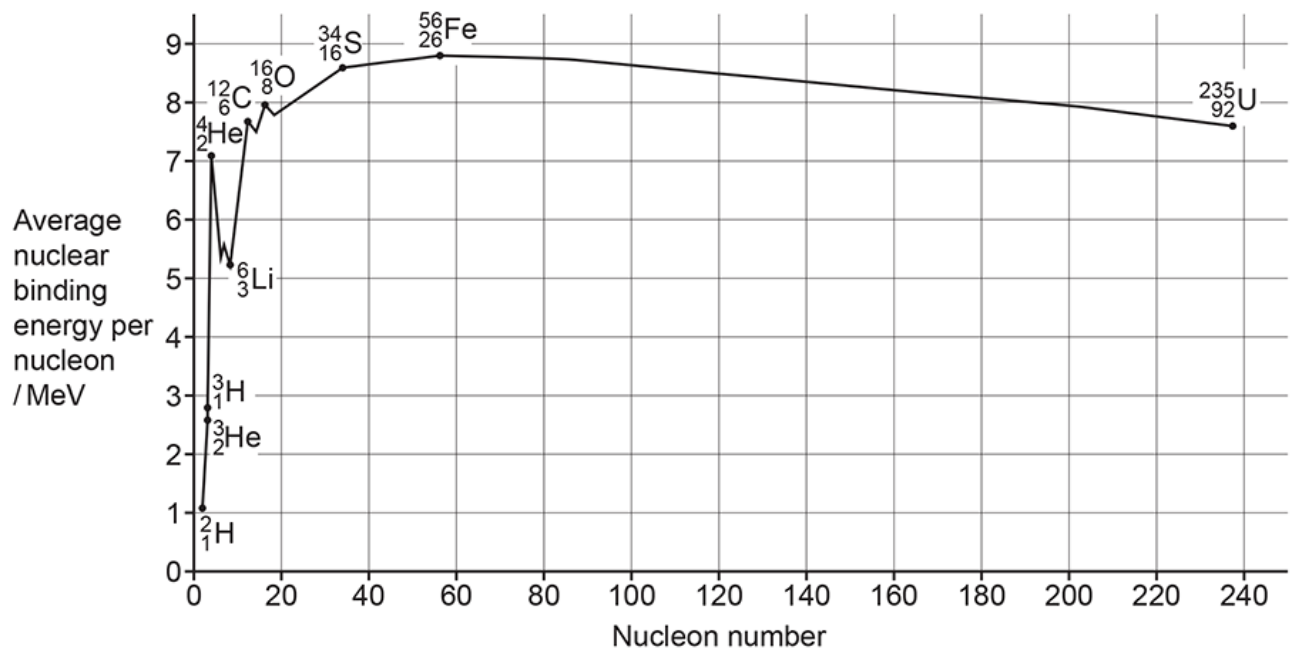
INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **96**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) The diagram below shows the average nuclear binding energy per nucleon for a number of different isotopes.



Explain what is meant by *nuclear binding energy* of a nucleus.

[1]

(b) Suggest why the ^1_1H isotope of hydrogen has **not** been included on the above diagram.

[1]

- (c) The main nuclear fusion reaction in the Sun is between nuclei of deuterium (${}^2_1\text{H}$) and tritium (${}^3_1\text{H}$).

This reaction can be written as shown below.



- i. Explain why isotopes with low mass numbers, such as hydrogen, are those which undergo nuclear fusion.

----- [1]

- ii. Use the diagram given at the start of this question to show that, for the reaction of deuterium and tritium, the energy released in each fusion event is approximately 3×10^{-12} J.

[3]

- iii. The Sun's mass decreases by 4.3×10^9 kg every second. Assume that the mass loss is only due to this reaction.

Calculate the number of fusion events per second occurring in the Sun.

number of fusion events per second = s^{-1} [2]

- 2 The energy released from the fusion of 1 kg of hydrogen is more than seven times the energy released by the fission of 1 kg of uranium.

Compare the practicalities of using nuclear fusion of hydrogen with using nuclear fission of uranium to meet our energy needs.

[4]

- 3(a) Radiographers commonly use molecules containing fluorine F-18 as tracers in positron emission tomography (PET) scanning.

Fluorine has a proton number of 9.

F-18 decays to oxygen (O) by β^+ decay.

Write the equation for the decay of a nucleus of F-18 using nuclear notation.

[2]

- (b) The β^+ particle (positron) produced travels only a short distance in the patient before it meets an electron and is annihilated.

Calculate the wavelength λ of gamma photons produced.

$\lambda = \dots\dots\dots\text{m}$ [3]

(c) X-rays and gamma-rays are produced by different physical processes.

Briefly describe both processes.

[2]

(d) F-18 has a half-life of 109.7 minutes.

Explain the advantage that this has for the patient but the disadvantage that this has for the radiographers.

[3]

4(a) Describe how fission of nuclei is induced and controlled in a nuclear reactor.

Show how fission leads to the release of large amounts of energy.

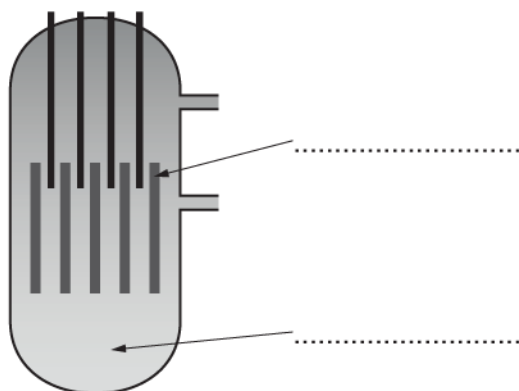
The following masses may be useful.

Particle	Mass / u
U-235 nucleus	235.04395
Ba-141 nucleus	140.91440
Kr-92 nucleus	91.92617
	1.00867

[6]

Handwriting practice lines consisting of 20 horizontal dashed lines.

(b) The diagram shows a simplified layout of a nuclear fission reactor used in a nuclear power station.



Complete the labels on the diagram

[2]

- 5 The mass of a proton is m_p , the mass of a neutron is m_n , and the mass of a hydrogen-3 (${}^3_1\text{H}$) nucleus is M . The speed of light in a vacuum is c .

Which expression is correct for the binding energy (B.E.) of the hydrogen-3 nucleus?

- A B.E. = $M \times c^2$
- B B.E. = $(m_n + m_p - M) \times c^2$
- C B.E. = $(m_n + 2m_p - M) \times c^2$
- D B.E. = $(2m_n + m_p - M) \times c^2$

Your answer

[1]

- 6 A gamma-ray photon of frequency 6.76×10^{22} Hz creates a particle-antiparticle pair. The particle-antiparticle pair have zero kinetic energy.

What is the mass of the particle?

- A 2.49×10^{-28} kg
- B 4.98×10^{-28} kg
- C 7.47×10^{-20} kg
- D 4.48×10^{-11} kg

Your answer

[1]

[2]

Here is some information and data on fission and fusion reactions.

- Describe the similarities and the differences between fission and fusion reactions.
- Explain with the help of calculations, which fuel produces more energy per kilogram.

[illegible]

Blank lined paper for writing.

Carbon-14 ($^{14}_6\text{C}$) is produced in the upper atmosphere of the Earth by collisions between nitrogen nuclei and fast-moving neutrons.

The nuclear transformation equation below shows the formation of a single carbon-14 nucleus.

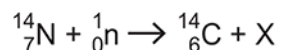


Fig. 22

- i. State the proton number of particle X.

proton number = [1]

- ii. Use the data below to determine the binding energy per nucleon of the $^{14}_6\text{C}$ nucleus.
Write your answer to 3 significant figures.

mass of neutron = 1.675×10^{-27} kg

mass of proton = 1.673×10^{-27} kg

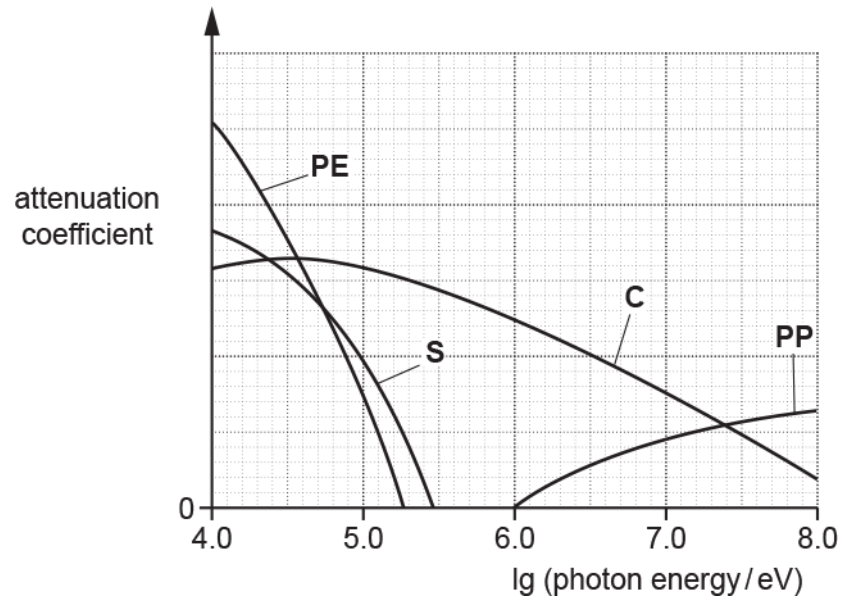
mass of $^{14}_6\text{C}$ nucleus = 14.000 u

1 u = 1.66×10^{-27} kg

binding energy per nucleon = J per nucleon [4]

9 X-ray photons interact with atoms.

The attenuation coefficient against $\lg(\text{photon energy})$ graphs for simple scattering (S), photoelectric effect (PE), Compton effect (C) and pair production (PP) are shown below.



With the help of a calculation, explain the minimum photon energy for pair production.

[3]

10 The total energy released in a single fusion reaction is 4.0 MeV.

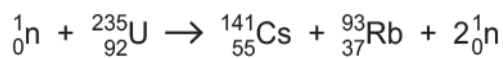
What is the change in mass in this fusion reaction?

- A 7.1×10^{-36} kg
- B 7.1×10^{-30} kg
- C 2.1×10^{-21} kg
- D 4.4×10^{-17} kg

Your answer

[1]

11 Inside a nuclear reactor, fission reactions are controlled and **chain reactions** are prevented. A typical fission reaction of the uranium-235 nucleus ($^{235}_{92}\text{U}$) is illustrated below.



The neutron triggering the fission reaction moves slowly. The neutrons produced in the fission reaction move fast.

i. Describe what is meant by **chain reaction**.

[2]

ii. Explain how chain reactions are prevented inside a nuclear reactor.

[2]

iii. The energy released in each fission reaction is equivalent to a decrease in mass of 0.19 u.
A fuel rod in a nuclear reactor contains 3.0% of uranium-235 by mass.

Estimate the total energy produced from 1.0 kg of fuel rod.

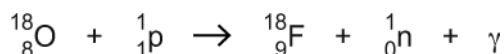
molar mass of uranium-235 = 0.235 kg mol⁻¹

1 u = 1.66 × 10⁻²⁷ kg

energy =J [4]

12

The nuclear reaction below shows how the isotope of fluorine-18 (¹⁸₉F) is made from the isotope of oxygen-18 (¹⁸₈O) .



The oxygen-18 nucleus is **stationary** and the proton has kinetic energy of 0.25 × 10⁻¹¹ J.

The binding energy of the ¹⁸₈O nucleus is 2.24×10⁻¹¹ J and the binding energy of the ¹⁸₉F nucleus is 2.20×10⁻¹¹ J. The proton and the neutron have zero binding energy.

- i. Explain why a high-speed proton is necessary to trigger the nuclear reaction shown above.

----- [2]

- ii. Estimate the minimum wavelength λ of the gamma ray photon (γ).

$\lambda =$ ----- m [3]

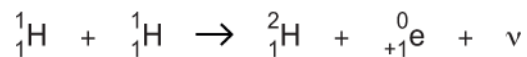
iii. Fluorine-18 is a positron emitter.

Name a medical imaging technique that uses fluorine-18 and state one benefit of the technique.

----- [2]

13(a) Stars produce energy by nuclear fusion.

One particular fusion reaction between two protons (${}^1_1\text{H}$) is shown below.



In this reaction 2.2 MeV of energy is released.

Only one of the particles shown in the reaction has binding energy.

Determine the binding energy per nucleon of this particle. Explain your answer.

----- [2]

(b) Explain why high temperatures are necessary for fusion reactions to occur in stars.

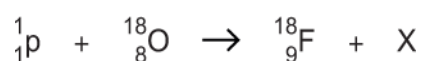
----- [2]

(c) In this reaction 2.2 MeV of energy is released.

A gamma photon in a star can spontaneously create an electron-positron pair.
Calculate the **maximum** wavelength of a gamma photon for this creation event.

maximum wavelength = _____ m [3]

- 14 A proton collides with a stationary oxygen-18 nucleus. The collision produces a fluorine-18 nucleus and particle X.



What is particle X?

- A neutron
- B proton
- C electron
- D positron

Your answer

[1]

The fusion of two ${}^3_1\text{H}$ nuclei produces a stable nucleus of ${}^4_2\text{He}$ and some fast-moving neutrons.

- i. Explain why the fusion of the ${}^3_1\text{H}$ nuclei must produce two neutrons.

----- [2]

- ii. The total energy released in this fusion reaction is 11 MeV. The binding energy per nucleon of the ${}^4_2\text{He}$ nucleus is 7.1 MeV.

Calculate in J the binding energy per nucleon of the ${}^3_1\text{H}$ nucleus.

binding energy per nucleon = _____ J [3]

16 Calculate the maximum wavelength of the X-rays for the pair production process.

maximum wavelength = _____ m [3]

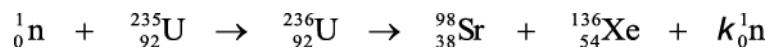
17(a) State what is meant by *induced nuclear fission*.

_____ [1]

(b) Explain the role of the moderator and the control rods in a nuclear reactor.

_____ [4]

(c) A possible fission reaction is



where k is the number of neutrons released in the reaction. The ${}_{92}^{236}\text{U}$ nucleus is very unstable.

i. State the number k of neutrons released in this reaction.

$k =$ _____ [1]

ii. State the binding energy of the released neutrons.

_____ [1]

iii. A nuclear reactor uses uranium-235 as fuel. The output power from the reactor is 1.0 GW. The mass of the ${}_{92}^{236}\text{U}$ nucleus is 236.053 u. The total mass of the fission products is 235.840 u.

Calculate the number of fission reactions per second.

number of reactions per second = _____ s^{-1} [4]

18 Fig. 22.1 shows the circular track of a positron moving in a uniform magnetic field.

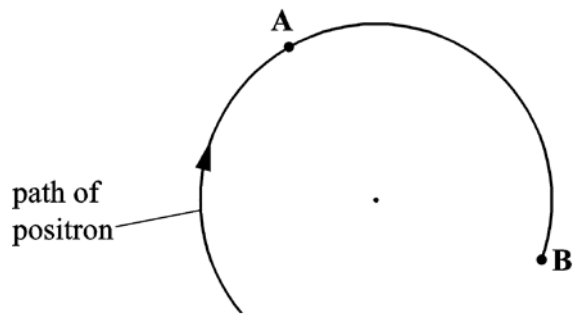


Fig. 22.1

The magnetic field is perpendicular to the plane of Fig. 22.1.

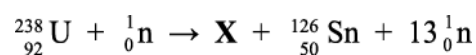
The speed of the positron is $5.0 \times 10^7 \text{ m s}^{-1}$ and the radius of the track is 0.018 m.

At point B the positron interacts with a stationary electron and they annihilate each other. The annihilation process produces two identical gamma photons travelling in opposite directions.

Calculate the wavelength of the gamma photons. Assume the kinetic energy of the positron is negligible.

wavelength = _____ m [3]

- 19 When a nucleus of uranium-238 absorbs a neutron, one combination of fission products can be tin-126 and a nucleus of element X. 13 neutrons are also emitted.



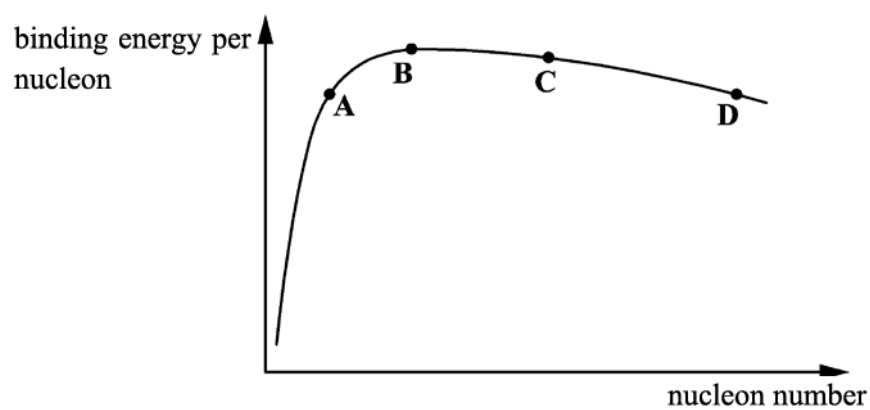
How many neutrons are there in the nucleus of element X?

- A. 30
- B. 42
- C. 58
- D. 100

Your answer

[1]

20 A graph of binding energy per nucleon against nucleon number is shown below.



Which nucleus, A, B, C, or D, shown on the graph has the largest magnitude of binding energy?

Your answer

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		(Minimum) energy required to separate the nucleons (to infinity)	B1	Allow protons and neutrons for nucleons Allow energy released when the nucleus forms from the nucleons Do not allow mathematical descriptions such as (mass defect $\times c^2$) <u>Examiner's Comments</u> A little less than half gave a suitable explanation for the binding energy; many incorrect responses were based on the idea that it was the energy holding the nucleus together. Several responses correctly stated the relationship between binding energy and mass defect although this did not answer the question.
	b		H-1 is a single proton / nucleon AND there is no binding energy / it is not bound to anything	B1	<u>Examiner's Comments</u> Around half of candidates gave a clear suggestion why hydrogen was not included. Many appreciated that the single nucleon had no binding energy although not all made a link. Several candidates described a hydrogen atom instead of the nucleus causing confusion over electron binding energy.
	c	i	The product has a higher binding energy (per nucleon)	B1	Allow the fusion of low Z number nuclei will result in a nucleus higher up the curve making it more stable Ignore statements regarding electrostatic repulsion <u>Examiner's Comments</u> This question was quite poorly answered with few candidates appreciating the idea that an increase in binding energy per nucleon would result in energy given out. Many candidates simply restated the question, by saying low mass would join to give higher mass or stated that iron was the most stable isotope.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	Energy before = $(1.1 \times 2) + (2.8 \times 3)$ (= 10.6 MeV) Energy after = 7.1×4 (= 28.4 MeV) Energy released = $28.4 - 10.6 = 17.8$ MeV $= 17.8 \times 10^6 \times 1.6 \times 10^{-19} = 2.8 \times 10^{-12}$ (J)	C1 C1 A1	Allow ± 0.1 MeV tolerance on binding energy values from graph Answer to at least 2sf (2.848) MAX value of energy within tolerances = 2.99×10^{-12} MIN value of energy within tolerances = 2.70×10^{-12} <u>Examiner's Comments</u> As always with a 'show that' question, it is vital to include as many steps as possible, the key to this question involved reading from the graph and then including the number of nucleons in the calculation. A fair number of candidates correctly read the point but did not multiply this with the nucleon number. There were relatively few misreads from the graph and the tolerance allowed some variation. Many simple (and incorrect) methods would give a value fairly close to 3×10^{-12} and so candidates would feel that they had correctly answered the question. Very few candidates struggled with the conversion from MeV to joules, although this may have been supported by having the power of ten given to them. It is also important to make sure that any value is given to a greater number of significant figures than that asked for, if appropriate. Around a third of candidates were able to obtain all 3 marks on what a potentially challenging calculation is.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	Energy released per second = $4.3 \times 10^9 \times (3 \times 10^8)^2 = 3.87 \times 10^{26}$ (J) No of reactions = $3.87 \times 10^{26} / 2.85 \times 10^{-12} = 1.4 \times 10^{38}$ (s⁻¹)	C1 A1	Alternative method: Mass per reaction = $2.85 \times 10^{-12} / (3 \times 10^8)^2 = 3.2 \times 10^{-29}$ (kg) No of reactions = $4.3 \times 10^9 / 3.2 \times 10^{-29} = 1.4 \times 10^{38}$ (s⁻¹) Allow use of 3×10^{-12} to give 1.29×10^{38} Allow use of (c)(ii) tolerance range for energy released value from to final answer in range 1.29 – 1.43 ($\times 10^{38}$) Correct to at least 2sf <u>Examiner's Comments</u> There are several routes to obtain the correct answer and credit is always given to any correct method. Neatly two thirds of candidates scored both marks on this question, even if they had not completed the previous part by using the 'show that' value. Candidates should always be aware that a 'show that' value is likely to be used in subsequent calculations.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2			Marks may be awarded for the following considerations only: technology, reaction, fuel, waste, safety. Technology: - Fission: technology already established / Fusion: technology not yet established Reaction: - Fission: chain reaction self-sustaining once initiated - Fusion: very high temperatures/pressures /amounts of energy required to initiate and continue reaction / challenges with storage and containment of plasma Fuel: - Fission: finite amount of uranium / needs to be enriched - Fusion: (unlimited) fuel available from (sea) water Waste: - Fission: Produces highly / high level / long half-life / long storage time radioactive waste - Fusion: Produces (effectively) no radioactive waste (products) / only radioactive material in contained in the reactor Safety: - Fission: Possibility of serious accident / leakage of radioactive material / "meltdown" AND/OR Fusion: very unlikely to produce serious accident (due to near instant shut down of reaction) / no leakage of radioactive material	4 x B1	ALLOW 1 mark for basic comparison within each consideration if no other marks are awarded. ALLOW hydrogen is easier to source (than uranium) for 1 mark ALLOW Fission produces radioactive waste and fusion does not for 1 mark. IGNORE nuclear for radioactive <u>Examiner's Comments</u> This question needed candidates to use appropriate scientific knowledge and understanding to discuss the differences between fission and fusion in terms of the practicalities. Many candidates were able to give good responses to several of the main points with careful comparisons. One of the key features of this question is to avoid social and economic arguments and base the points on ideas from the course. It is not a 'compare and contrast' question, but many of the marking points can be gained that way. A relatively small number of candidates gained full marks on this, as may be expected, however a good fraction gained 2 or more marks. Responses which could not be credited were simple statements like 'safer', 'easier to use', 'no toxic waste' and 'renewable'. Examiners will have worked strictly to the mark scheme, and so given marks consistently.
			Total	4	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
3	a	${}^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^0_1\beta^+ (+\nu)$	C1 A1	Correct fluorine isotope Correct equation ALLOW numbers written to right of symbol ALLOW $e / e^+ / \bar{e} / \beta$ for positron symbol with correct numbers IGNORE gamma in products ALLOW ${}^0_0\nu$ for neutrino ALLOW ν_e for neutrino Examiner's Comments Most candidates will have gained the first mark for the correct isotope of fluorine. Those who didn't likely reversed the positions of the nucleon and proton number. It was noticeable that a large number of candidates were unable to balance the equation and while they had the correct values for the positron, the oxygen isotope was incorrect.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$E = mc^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2$ $= 8.2 \times 10^{-14} \text{ (J)}$ $E = hc / \lambda = 8.2 \times 10^{-14} = 6.63 \times 10^{-34} \times 3 \times 10^8 / \lambda$ $\lambda = 3.00 \times 10^8 \div 1.2367 \times 10^{20} = 2.4 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Working leading to evaluation of E ALLOW factor of 2 leading to 1.6×10^{-13} (J) NOT use of $\lambda = h/mv$ where v is 3×10^8 alone XP ALLOW use of $\lambda = hc/mc^2$ or $\lambda = h/mc$ ALLOW Use of electron mass = 0.511 MeV giving 8.2×10^{-14} (J) Correct to at least 2 significant figures Answer of 4.9×10^{-12} or 1.2×10^{-12} allow 2/3 due to incorrect accounting for factor of 2 <u>Examiner's Comments</u> Well over half of the candidates were able to score 2 or more marks on this question. A great deal scored 2 marks due to the inclusion of a factor of two, which was then not removed (or vice versa). Some candidates used the formula for kinetic energy – which by coincidence may give the correct answer – however this was not credited, nor was the more common use of the de Broglie formula. Examiners sometimes had a difficult decision on this question, whether the response was a physics error or not, and if in doubt a candidate would be awarded the marks.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	c	X-rays formed when <u>electrons</u> (in an atom) de-excite Gamma rays come from the decay of <u>nuclei</u> (in unstable isotopes)	B1 B1	ALLOW X-rays may be produced by acceleration / deceleration of (fast moving) electrons / X-rays are produced when (fast moving) electrons are incident on a metal target / X-rays may be produced when electrons lose kinetic energy ALLOW gamma rays come from (the decay of) radioactive <u>nuclei</u> / gamma rays come from the <u>nucleus</u> of unstable atoms / gamma rays come from the de-excitation of <u>nuclei</u> / gamma rays come from <u>annihilation</u> of particle-antiparticle (pairs) / IGNORE gamma from fission <u>Examiner's Comments</u> The main principle behind this question was to distinguish between the X-ray production by electrons and gamma ray production by nuclei. Fairly specific details were required for each, although a wide range of answers were accepted. This question discriminated well among the candidates with roughly equal fractions getting 0, 1 and 2 marks.
	d	The half-life is short Advantage: Exposure of the patient to <u>ionising</u> radiation is kept as low as possible. Disadvantage: (Radiographers have a) short time to scan/diagnose the patient	B1 B1 B1	ALLOW activity is high IGNORE it decays quickly ALLOW F18 has to be manufactured on site before use / high activity means exposure is high during handling IGNORE short time to treat the patient <u>Examiner's Comments</u> Although apparently a simple set of ideas, the detail required for this question meant that only a small fraction of candidates were able to gain full marks. It was necessary to state that the half-life was (relatively) short, which many of the stronger candidates did not do. The advantage to the patient requires the use of the term 'ionising' as this is the fundamental issue with the radiation. Similarly, candidates had to make it clear that the disadvantage was due to the short time to carry out the scan rather than just a statement of needing to work quickly.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	10	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
4	a	Level 3 (5–6 marks) Clear description and full correct calculation. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description or full correct calculation. or Some description and some calculation <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or Limited calculation <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark <i>No response or no response worthy of credit.</i>	6 x B1	Use level of response annotations in RM Assessor Indicative scientific points may include: Description • Absorption of neutron by U-235 • U-235 nuclei become unstable U236 nuclei • U-236 undergoes fission • Resulting in smaller nuclei daughter products and additional fast neutrons • Moderator slows neutrons so that they can be absorbed by U-235 nuclei • Process repeats to form a chain reaction • Rate of chain reaction controlled by raising or lowering control rods • Which absorb excess neutrons • To allow one neutron per fission to cause further fission Calculation Use of ${}_0^1\text{n} + {}_{92}^{235}\text{U} \rightarrow {}_{36}^{92}\text{Kr} + {}_{56}^{141}\text{Ba} + 3{}_0^1\text{n}$ Mass of U236 = 235.04395 + 1.00867 = 236.05262u Total mass of fission products = 140.91440 + 91.92617 + (3 x 1.00867) = 235.86658u Difference = 0.18604 u Mass defect = 0.18604 x 1.661 x 10⁻²⁷ = 3.0901244 x 10⁻²⁸ kg Energy equivalence per fission event = 3.0901244 x 10⁻²⁸ x (3 x 10⁸)² = 2.78111x10⁻¹¹J Alternative: Energy equivalence per fission in MeV = 174 MeV Examiner's Comments This is the second level of response (LoR) question in this paper, and was designed to assess nuclear physics, specifically Sections 6.4.4 (a), (b), (g) and (h). In

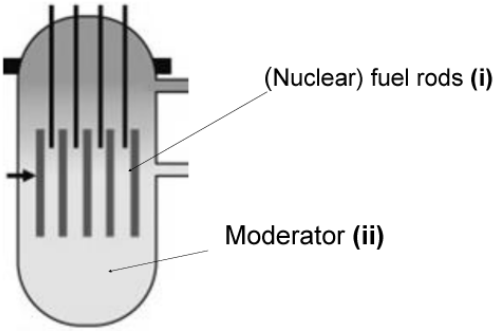
Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			particular, there are two parts (description and calculation) both of which need to be answered to obtain a Level 3. As previous, there are no specific marking points and examiners use a 'best fit' approach to the marking. Indicative scientific points are given in the mark scheme as a guidance to what to expect, but not all need to be satisfied for full credit, however for full credit for the calculation, the energy released per fission is the minimum end point. Incorrect physics can be expected to be penalised, however. Candidates should always be encouraged to structure their answers carefully and in calculations aim to give explanation of each stage rather than just produce a number. The two separate sections can be separated into a description of the induction and control, and calculation of the energy released. The key points in the description that examiners were looking for were: - idea of induced fission by a thermal neutron– the chain reaction leading to more neutrons released - a description of the role of the moderator – a description of the role of the control rods. The key points in the calculation that examiners were looking for were: - calculation of the mass before fission – calculation of the mass after the fission – difference between these two masses – calculation of the energy equivalent (hence released) from this mass difference. The descriptions of the induction and control varied hugely both in quality and quantity. Many candidates did not start from U-235 but went straight into the fission. Most candidates were aware that extra neutrons were released in fission and had a good idea of the absorption of the extra neutrons by the control rods, but there was often confusion about the role of the moderator (by explaining that it simply 'controlled the reaction') and a significant number did not appreciate why a slower neutron was needed. However, there were many excellent explanations covering all of the indicative points and going beyond. The calculations were attempted well by many candidates and obtaining the energy

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			equivalence of one fission was a suitable point to stop the calculation. A comment supporting the size of the energy compared to this number of fissions per second helped to confirm the large release of energy. Several candidates took this further and calculated the energy per kg, showing how large this was. The main source of error was in not accounting for the mass of the initial neutron in the reaction and so the energy calculated was one 'neutron mass equivalence' too high. Several candidates ended their calculation with a mass difference, or even attempted to calculate the mass defect in the formation of U-235, but this led nowhere. There was a fair number of candidates who made no attempt on the calculation despite having given a good description and it can be assumed that they hadn't fully read what they were asked to do. Similarly there were some excellent calculations but no with no description. Candidates are always encouraged to consider both parts of any LoR question. Exemplar 3 $\begin{aligned} & \text{U-235} \rightarrow \text{Ba-141} + \text{Kr-92} + 3\text{n} \\ & (235.043929) \quad (140.91441) + (91.92615) + (3 \times 1.008665) \\ & = 235.86658 \end{aligned}$ $\begin{aligned} \Delta m &= 235.043929 - 235.86658 = -0.822651 \\ &= -0.822651 \times 1.66054 \times 10^{-27} = -1.366 \times 10^{-27} \text{ kg} \end{aligned}$ $\begin{aligned} \Delta E &= \Delta mc^2 \\ &= 1.366 \times 10^{-27} \times (3 \times 10^8)^2 = 1.23 \times 10^{-10} \text{ J per reaction} \end{aligned}$ The moderator A fission reaction is induced by a neutron being absorbed by a radioactive nucleus, which is split into two lighter nuclei and more neutrons. Energy is released. Neutrons are then absorbed by other radioactive nuclei causing a chain reaction. The moderator slows down the fast moving neutrons released in the reactions into thermal neutrons, as they are more likely to be captured by the reaction. Control rods absorb the thermal neutrons leaving on average one left from each reaction so the reaction is controlled. This is an example of a Level 2 response to a LoR question. The candidate has given a clear calculation with a good structure but has forgotten to add the neutron mass to the uranium mass to give the initial mass. The remainder of the calculation is correct, giving a final value for the energy per fission which is incorrect by a value of the neutron mass-energy.

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					The explanation of the induction and control is good, if a little brief and lacking in real detail. On the holistic approach taken to the LoR questions, this is a clear Level 2. The communication is good and the setting out of the response is clear.
	b		 (Nuclear) fuel rods (i) Moderator (ii)	B1 B1	ALLOW any correctly named fuel e.g. uranium (oxide) for (i) ALLOW any correctly named moderator e.g. (heavy) water for (ii) ALLOW coolant for (ii) ALLOW any correctly named coolant (e.g. CO₂) for (ii) Examiner's Comments Many candidates were able to demonstrate their knowledge of the internal structure of a fission reactor and correctly label the components or the material they are made from. Simple answers such as 'reactor core' or 'fluid' are not sufficient at this level.
			Total	8	
5			D	1	Examiner's Comments This question can correctly be answered by over three quarters of the candidates. Most appeared to be able to evaluate the composition of the nucleus correctly and then work forward from that. A was the most common incorrect response, presumably as less successful responses were able to recognise a variation of $E = mc^2$.
			Total	1	
6			A	1	Examiner's Comments This question was answered correctly by approximately half of the candidates. Most showed written working, often as $hf = 2mc^2$ or the equivalent in numerical form. B was by far the most common incorrect response, and it is clear from candidates working that the factor of 2 had simply been missed out.

Mark Scheme

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			Total	1	

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Question		Answer/Indicative content	Marks	Guidance
7	a	Control rods: absorb the <u>neutrons</u> (without further fission) Moderator: Slow down the <u>neutrons</u> / decrease KE of <u>neutrons</u>	B1 B1	Not collide for absorb Examiner's Comments For this question, the candidates need to explain the role of these components in terms of their interactions with neutrons and those who did not mention neutrons at all in their responses could not score any marks. Many candidates went beyond what was required and explained what effect this has on the reactor, such as controlling the rate of reaction. In general, this question was not answered well.  Misconception Many candidates gave vague statements regarding the function of these components rather than an explanation.
	b	Level 3 (5–6 marks) Clear description and clear calculations of energy per kg <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description OR Clear calculations of energy per kg OR Some description and some calculations <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description OR Limited calculations <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks No response or no response worthy of credit	B1×6	Indicative scientific points may include: Description • Energy is produced in both reactions • More energy produced (per reaction) in fission • The (total) binding energy of 'products' is greater • In fusion, nuclei repel (each other) • Fusion requires high temperatures / high KE • Fission reactions are triggered by (slow-)neutrons • Chain reaction possible in fission Calculations • 1 kg of uranium has 4.26 mols / 2.56×10^{24} nuclei • 1 kg of deuterium has 500 mol / 3.01×10^{26} nuclei / 1.50×10^{26} 'reactions' • $200 \text{ MeV} = 3.2 \times 10^{-11} \text{ J}$ • $4 \text{ MeV} = 6.4 \times 10^{-13} \text{ J}$ • Uranium: $\sim 10^{14} \text{ (J kg}^{-1}\text{)}$ (actual value 8.2×10^{13}) • Deuterium: $\sim 10^{14} \text{ (J kg}^{-1}\text{)}$ (actual value 9.6×10^{13}) • The energy per kg is roughly the same Examiner's Comments

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					This is the second LoR question. This is designed to assess knowledge of the two nuclear energy reactions and to calculate energy release using some given data. The differences between the fission and fusion reactions were generally well answered although many candidates explained differences in design, operation and waste more than the reactions. The similarities were often not as clear however several candidates gave excellent responses in terms of binding energies and mass differences. Candidates were also expected to complete a calculation to show which produces more energy output per kilogram. This is challenging calculation to follow through fully, but most candidates were able to make some attempt, even if it was only converting MeV to J. Only better candidates realised 2 nuclei of deuterium were used for one fusion reaction. While a small number of candidates did correctly calculate the energy per kilogram, they tended to state that fusion produced more energy rather than a feeling that they are basically equivalent. As usual with LoR questions, a holistic approach is taken to the marking and candidates can access higher levels without necessarily reaching all the marking points. Even so, relatively few candidates were able to access Level 3, generally due to poor calculations and/or descriptions.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8		i	1	B1	<u>Examiner's Comments</u> This question was correctly answered by the vast majority of candidates.
		ii	Either: mass of nucleus $14.000 \times 1.66 \times 10^{-27}$ $(= 2.324 \times 10^{-26} \text{ kg})$ Or: mass of nucleons $= 8 \times 1.675 \times 10^{-27} + 6 \times 1.673 \times 10^{-27}$ $(= 2.3438 \times 10^{-26} \text{ kg})$ $(\Delta m =) 2.3438 \times 10^{-26} - 2.324 \times 10^{-26} =$ $(1.98 \times 10^{-28} \text{ kg})$ $(\Delta E =) 1.98 \times 10^{-28} \times (3.00 \times 10^8)^2$ (BE per nucleon $=) 1.782 \times 10^{-11}/14$ binding energy per nucleon $= 1.27 \times 10^{-12}$ (J per nucleon)	C1 C1 C1 A1	$\Delta m = 1.9262 \times 10^{-28} \text{ kg}$ Ignore sign throughout $\Delta E = 1.782 \times 10^{-11} \text{ J}$ Allow for any mass difference $\times (3.00 \times 10^8)^2$ Note A mark for correct answer to 3sf only <u>Examiner's Comments</u> This final calculation required some careful structure and several stages. An encouraging number were able to work through the solution to its conclusion. Some rounded intermediate calculations too early and so lost the final 3 significant figures mark. Several candidates also missed the division by the nucleon number, either as a slip or perhaps they did not appreciate that this was what was required. Even the weakest candidates realised the need to apply $E = mc^2$, but would only gain credit here if they had calculated a mass difference. Some candidates also miscalculated the number of protons and neutrons in the carbon nucleus, which meant that they were limited to a maximum of 2 marks.
			Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9			(energy =) $9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$ (energy =) $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 / 1.60 \times 10^{-19}$ $\lg 1.0(2) \times 10^6 = 6$ (as on graph) OR (energy =) 1.0×10^6 (eV) or $\lg 1.0 \times 10^6 = 6$ (from graph) (energy =) 1.6×10^{-13} J and evidence of mc^2 $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 \approx 1.6 \times 10^{-13}$	B1 B1 B1 B1 B1 B1	Note this is 8.2×10^{-14} (J) Note this is $1.0(2) \times 10^6$ eV Note this can be shown in a variety of ways
			Total	3	
10			B	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11		i	More neutrons produced (from each fission reaction)	B1	<u>Examiner's Comments</u> Most candidates scored 1 mark for the general idea of a chain reaction, but the important role played by the neutrons was often omitted in the descriptions. Only a small number of candidates misunderstood fission as a reaction in which the Cs and Rb nuclei themselves were responsible for triggering subsequent reactions of the uranium nuclei.
			Go on to produce further (fission) reactions / splitting (of nuclei) / energy	B1	
		ii	Control rod(s) used	B1	Allow boron / cadmium / indium / silver Not moderator Not neutrons slowed down and/or stopped <u>Examiner's Comments</u> The mechanism of preventing uncontrolled chain reaction within a nuclear reactor was generally well understood. Having given perfect answers with control rods absorbing the excess neutrons, a significant number of candidates confusing their answers by also mentioning the moderator. It many cases it was impossible for examiners to decide from the candidates response if the control rods, or the moderators, were responsible for preventing chain reactions. Some candidates mentioned 'boron rods', this was an acceptable alternative for the 'control rods'.
			These absorb the neutrons (without fission)	B1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	$(\Delta m =) 0.190 \times 1.66 \times 10^{-27}$ or 3.15×10^{-28} (kg) $(\Delta E =) 0.190 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ or 2.84×10^{-11} (J) $\frac{1.00}{0.235} \times 6.02 \times 10^{23}$ or 2.56×10^{24} (energy = $0.03 \times 2.56 \times 10^{24} \times 2.84 \times 10^{-11}$) energy = 2.2×10^{12} (J)	C1 C1 C1 A1	Note the 3.0% can be done at any stage, allow other correct methods Allow the use of 1.67×10^{-27} Allow ECF from 1.66×10^{-27} omitted Note 7.69×10^{22} will score this C1 mark; 3.0% included Allow 3 marks for 7.3×10^{13} (J); 3.0% omitted Allow 3 marks for 1.3×10^{39} (J); 1.66×10^{-27} omitted Examiner's Comments This proved to be an excellent discriminator with top end candidates showing excellent skills to get to the correct answer of 2.2×10^{12} J. The majority of the candidates correctly converted the 0.19u into kilograms, and then successfully used Einstein's mass-energy equation to calculate the equivalent energy of 2.8×10^{-11} J. The main obstacle in this question was the determination of the number of uranium nuclei in the fuel rods. Avogadro constant, given in the data booklet, was either omitted or the incorrect mass used to determine the number of uranium nuclei.  Misconception There were some missed opportunities, with some candidates making the following mistakes when determining the number of uranium nuclei in the 1.0 kg fuel rod. - Using $0.235 \times N_A$ to calculate the number of uranium nuclei. - Using the rest masses of neutrons and protons. - Omitting the 3.0%.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
12		i	Proton is repelled (by nucleus) (High-speed) proton can get close to (oxygen) nucleus	B1 B1	Allow 'proton can experience the strong (nuclear) force' Not 'collide / hit nucleus'
		ii	$E = [0.25 - (2.24 - 2.20)] \times 10^{-11} \text{ (J) or } 0.21 \times 10^{-11} \text{ (J)}$ $\lambda = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{0.21 \times 10^{-11}} \text{ (Any subject)}$ $\lambda = 9.5 \times 10^{-14} \text{ (m)}$	C1 C1 A1	Allow 2 marks for 6.9×10^{-14} ; $E = 0.29 \times 10^{-11}$ used Allow 1 mark for a value correctly calculated based on any other incorrect value for E (e.g. $8(.0) \times 10^{-14}$ for $E = 0.25 \times 10^{-11}$ and $5(.0) \times 10^{-13}$ for $E = 0.04 \times 10^{-11}$)
		iii	Used in PET (scans) Any one from: Used to diagnose function of organ / brain / body Detection of cancer / tumour Non-invasive / no surgery / no infection 3D (image)	M1 A1	Enter text here.
			Total	7	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
13	a	${}^2_1\text{H}$ has two nucleons binding energy per nucleon = 1.1 <u>MeV</u> (per nucleon)	B1 B1	Allow $1.76 \times 10^{-13} \text{ J}$ (per nucleon) Examiner's Comment About half of the candidates scored one or two marks. Some of the answers were concise with just '<i>1.1 MeV per nucleon</i>' because ${}^2_1\text{H}$ has two nucleons' but some were simply incorrect with an attempt to answer the question using $\Delta E = \Delta mc^2$ and the rest masses of the proton and neutron. Weaker candidates misunderstood the terms 'binding energy' and 'binding energy per nucleon'.
	b	The <u>protons</u> / <u>nuclei</u> repel each other (At high temperature) particles have more <u>KE</u> and hence can get <u>close</u> (enough to fuse)	B1 B1	Not atoms / particles Allow 'enough <u>KE</u> to get close' Not atoms or ions Examiner's Comment Fusion is possible in stars because the higher temperatures ensure that nuclei have large enough kinetic energy for the most energetic ones to get close enough for the attractive strong nuclear force to trigger the reactions. Many candidates did realise that higher temperatures meant greater kinetic energy but some answers were spoilt for either stating that atoms were fusing or mentioning that '<i>nuclei overcame electrical forces</i>' without any further explanation. The superficiality of many answers prevented candidates from picking up marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		$E = hc/\lambda$ and $E = me^2$ or $E = 2 \times mc^2$ $\lambda = \frac{6.63 \times 10^{-34}}{2 \times 9.11 \times 10^{-31} \times 3.0 \times 10^8}$ maximum wavelength = 1.2×10^{-12} (m)	C1 C1 A1	Allow $hc/\lambda = 2mc^2$ with or without the factor of 2 Note: The mass must be $2m_e$ to score this and the next mark Not de Broglie equation $\lambda = h/mv$ with speed of c; which gives 2.4×10^{-12} (m) Allow 2 marks for 6.6×10^{-16} (m); mass of neutron or proton used instead Allow the following marks for 1.02 MeV recalled: $E = 1.63 \times 10^{-13}$ (J) C1 $\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.63 \times 10^{-13}}$ C1 maximum wavelength = 1.2×10^{-12} (m) A1 Examiner's Comment This was another question that favoured the top-end candidates. The answers from them showed excellent understanding of pair production. A small number of candidates correctly recalled the 1.02 MeV energy required to produce an electron-positron pair. Full credit was given if this led to the correct answer of 1.2×10^{-12} m for the gamma photon. A disappointing number of candidates used 2.2 MeV from (a) and the equation $\lambda = hc/E$ to calculate the maximum wavelength. There was no credit for this incorrect approach.
			Total	7	
14			A	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
15		i	total nucleon number after fusion = $3 + 3 - 4 = 2$	M1	Allow other correct methods
		i	total proton number after fusion = $1 + 1 - 2 = 0$	M1	
		i	(Hence it must be 2 neutrons ${}_0^1\text{n}$ after the fusion reaction)	A0	
		ii	(BE of neutron(s) = 0 and BE of ${}_2^4\text{He} = 28.4 \text{ MeV}$ BE of ${}_1^3\text{H}$ nucleus = $\frac{1}{2} \times (28.4 - 11) = 8.7 \text{ (MeV)}$	C1	
		ii	BE per nucleon = $8.7/3 = 2.9 \text{ (MeV)}$ BE per nucleon = $2.9 \times 10^6 \times 1.60 \times 10^{-19}$	C1	
		ii	BE per nucleon = $4.6 \times 10^{-13} \text{ (J)}$	A1	
			Total	5	
16			$\frac{hc}{\lambda} = 2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$	C1	
			$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2}$	C1	
			$\lambda = 1.2 \times 10^{-12} \text{ (m)}$	A1	
			Total	3	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
17	a		The splitting of a (uranium) nucleus as a neutron is absorbed (into two fragment nuclei and neutrons).	B1	
	b		The moderator slows down the fast-moving neutrons. The neutrons lose significant amount of their kinetic energy when colliding with moderator nuclei. or The moderator does not absorb the neutrons. The control rods absorb the neutrons. The rate of fission reactions is less / reduced.	B1 B1 B1 B1	
	c	i	2	B1	
		ii	Zero	B1	
		iii	$\Delta m = 236.053 - 235.840 = 0.213 \text{ u}$	C1	
		iii	$\Delta E = [0.213 \times 1.661 \times 10^{-27}] \times (3.0 \times 10^8)^2 = 3.184 \times 10^{-11} \text{ (J)}$	C1	
		iii	number of reactions per second = $10^9 / 3.184 \times 10^{-11}$	C1	
		iii	number of reactions per second = $3.1 \times 10^{19} \text{ (s}^{-1}\text{)}$	A1	
			Total	11	
18			energy of <u>two</u> photons = $2 \times mc^2$ or $2 \times \frac{hc}{\lambda} = 2 \times mc^2$ $\lambda = \frac{h}{mc} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 3.0 \times 10^8}$ wavelength = $2.4 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Correct use of $\frac{hc}{\lambda} = mc^2$
			Total	3	
19			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
20			D	1	
			Total	1	